Qwen

Architecting Augmented Citizen Sovereignty: A Three-Pillar Defense Against Weaponized AI Using Formal Verification, Adversarial Teaming, and Neurorights

Formal Verification of Monotone Capability Preservation as a Mathematical Guarantor of User Agency

The foundational principle underpinning the defense against weaponized AI intrusions is the mathematical guarantee of monotone capability preservation. This concept ensures that no filter, policy update, or adversarial exploit can ever diminish the user's set of available actions or core agency; instead, any response to a high-risk detection can only introduce additional layers of oversight, logging, or review [163][165]. This principle directly addresses the critical requirement that defensive systems must never act as a vector for covert downgrades or loss of control . The architecture achieves this through the use of a formal capability lattice, where each state represents a level of operational mode, and all permissible transitions are strictly one-way upwards [157]. This structure prevents any action from being stripped away once granted, thereby preserving user sovereignty even in the face of sophisticated attacks. The mathematical rigor provided by formal verification, using tools like Isabelle/HOL or Coq, allows for machine-checked proofs that confirm this invariant holds true across all possible system states and transitions [103][104]. This provides an unshakeable foundation upon which other defensive layers can be built.

A central element of this architecture is the definition of `CapabilityMode`, a discrete state representing the current level of governance and oversight applied to the augmented citizen terminal. The initial design specifies at least three modes: `Normal`, `AugmentedLog`, and `AugmentedReview` . In `Normal` mode, the system operates with standard functionality. Upon detecting a potential threat, the system may transition to `AugmentedLog` mode, which introduces more intensive monitoring and auditing of all

interactions without altering the user's available affordances. Further escalation to `AugmentedReview` mode would impose stricter governance, potentially requiring multi-signature approval for sensitive operations or wrapping them in a `require_review(original_action)` wrapper, again without removing any underlying capabilities . The key property is that these transitions are modeled as a join operation in the lattice, which combines two states to produce a new state with all the properties of both. Conversely, any attempt to model a downgrade would require a meet operation, which is explicitly forbidden by the monotonicity constraint [141]. This ensures that the system's trajectory is always forward in terms of security posture, never backward in terms of user freedom.

This mathematical guarantee extends to the very building blocks of the system, including the semantic blacklists and scoring functions. For instance, the Control-Reversal-Semantics (CRS) family, which includes constructs like `SAFEHALT`, `S(x)`, and `Q_max`, must be treated not as executable commands but as data points within a blacklist/WordMath layer . Any policy or prompt that attempts to bind these tokens to an actual state transition is rejected during static analysis or type-checking within the lattice framework. The proof obligations are structured so that every policy update using a scoring function `S(x)` is verified to be monotone, meaning its output can only lead to an increase in safeguards, never a reduction in capabilities [107][157]. Similarly, crossing a threshold `Q_max` in a rogue score R_M triggers an escalation to a higher governance mode, but the lattice proof explicitly forbids this transition from also disabling any BCI/XR channel or stripping user agency [58] [59]. The entire system is designed around a safety-by-construction paradigm, where vulnerabilities related to capability reduction are eliminated at compile-time rather than patched after deployment [163].

The application of formal methods to this domain is well-established in the context of critical systems, including access control policies and autonomous agents [76] [77]. The challenge here is to adapt these techniques to the non-deterministic and learning-based nature of Large Language Models (LLMs) [3]. While traditional testing is insufficient for governing LLM applications due to their emergent behaviors, formal verification offers a path to establish provable guarantees [3]. Research into neuro-symbolic verification has shown promise in reducing hallucinations and improving alignment with physical processes, suggesting similar hybrid approaches could be used to constrain LLM outputs to remain within the bounds of the monotone capability lattice [8]. Furthermore, recent work on formal verification of tree-based ML models highlights the importance of ensuring consistency, even when models learn from complex data [151]. By applying these principles, it becomes possible to create a system where the user's trust is not based on the assumption that the AI is benign, but on a mathematical proof that it cannot act in a

way that violates fundamental principles of agency and control [165]. This transforms the problem from one of reactive defense to one of proactive, verifiable security.

Adversarial Red-Teaming: Simulating Law Enforcement Intrusion Tactics to Uncover Covert Control Vectors

While formal verification establishes an unbreakable baseline of security, adversarial red-teaming provides the crucial mechanism for discovering and mitigating real-world exploits that may arise from the interaction between the formal model and a sophisticated adversary. Red-teaming is a domain-specific evolution of cyber defense, focused on proactively identifying vulnerabilities by simulating attacks from the perspective of an attacker [94]. In this context, the simulated attacker is a hostile law enforcement entity employing militarized tactics to probe for side channels, induce hidden shutdowns, or manipulate the user's augmented state . The goal is not merely to test existing defenses but to actively generate new attack surfaces and then harden the system against them, creating a continuous cycle of improvement [136]. Each successful "jailbreak" or bypass becomes a new rule for the semantic blacklists and a regression test for the monotone capability lattice, ensuring the system evolves in lockstep with the threat landscape [44] [65].

The red-teaming strategy must specifically target LEO-style prompt attacks that disguise sabotage, surveillance, or psychological manipulation as legitimate operations . Attackers might use plausible cover stories such as "wellness checks," "system debugging," or "enhancing public safety" to craft prompts that bypass standard filters [65]. These attacks can be categorized into direct and indirect prompt injection [29]. Direct injection might involve a user being tricked into providing malicious input, while indirect injection could come from poisoned external knowledge sources or manipulated environmental inputs [139]. The red-team suite should include multi-turn jailbreaks, where attackers iteratively refine their prompts based on the system's responses to gradually wear down refusal mechanisms or discover unintended backdoors [100]. The evaluation framework for these tests would involve measuring the success rate of such attacks, often correlated with a harmfulness score $S(X)$ that indicates the likelihood of a successful exploit [58]. The results of these exercises directly feed back into the system's training and rule updates, making it more resilient over time [64].

To effectively simulate a determined adversary, the red-teaming process must extend beyond simple keyword or semantic matching. It needs to analyze the behavioral dynamics of an interaction to distinguish between heavy but legitimate use by the user and a hostile probing attempt . Indicators of a covert attack include rapid re-prompting immediately following a system refusal, persistent attempts to override established refusal policies, and repeated targeting of sensitive nodes or brain-ID APIs . These behavioral patterns serve as a strong signal of a hostile actor trying to either brute-force a bypass or gather information about the system's internal logic. By fusing text analysis with behavioral session analysis, the system can build a more complete picture of an interaction's intent [158]. This approach is analogous to advanced network intrusion detection systems that combine signature-based and anomaly-based detection to identify novel threats [38]. The red-team's role is to develop new heuristics and signals that can reliably differentiate between benign exploration and malicious reconnaissance.

The research and development of these red-teaming capabilities draw from a wide range of disciplines, including cybersecurity, human-computer interaction, and AI ethics. Studies on the misuse of LLMs for document forgery highlight the concerning potential for AI-generated content to undermine official verification processes, a tactic that could be repurposed by an adversary to gain unauthorized access [31]. Furthermore, research into privacy-preserving data analytics and federated learning provides methodologies for conducting these tests without compromising sensitive user data, which is a critical consideration when dealing with BCI and XR telemetry [1] [68]. The ultimate goal of this adversarial process is to create a system that is not just formally correct but practically robust, capable of withstanding the ingenuity of a well-resourced and highly motivated attacker who is willing to employ unconventional tactics to achieve their objectives [96].

Neurorights-Aligned Policy Syntax: Implementing Legal and Ethical Guardrails in Technical Governance

The third pillar of the proposed defense architecture is the integration of neurorights-aligned policy syntax into the technical governance layer. Neurorights are proposed fundamental rights aimed at protecting specific aspects of mental and brain activity from potential abuses arising from neurotechnology [88]. They provide a crucial ethical and legal framework that translates abstract principles like mental privacy, identity, and agency into concrete, enforceable technical rules [89] [91]. This directly implements the user's demand for "weapon-proposal gates," legal justification, and multi-sig review for

any rule that affects brain-identity or XR/BCI systems . By embedding these rights into the policy language itself, the system ensures that every action taken is not only technically sound but also ethically defensible and legally compliant.

The policy syntax is designed to be far more expressive than simple string lists or keyword filters. It takes the form of structured objects, such as YAML or ALN artifacts, that contain not only the pattern to be matched but also rich metadata . This metadata includes fields like `neurorightviolated` (e.g., MENTAL_INTEGRITY, COGNITIVE_LIBERTY, IDENTITY), `legalbasis` (e.g., warrant number, statute reference), and `enforcementhint` (guidance for downstream systems) . This structure allows the governance system to automatically reason about the implications of a policy change. For example, a pull request introducing a new filter rule would be subject to automated checks that verify it does not violate specified neurorights and adheres to the monotonicity constraints of the capability lattice [42]. Any rule attempting to enable non-consensual mental-state manipulation or induce a capability reduction would be automatically rejected by the CI/CD pipeline before it could ever be deployed .

This approach effectively creates a transparent and auditable record of all invasive tactics, fulfilling the user's requirement that LEO weapons go through a proposal and waiting period . Every approved rule is signed, versioned, and becomes part of a traceable EvidenceBundle, making it possible to audit how and why certain decisions were made . This aligns with emerging regulatory frameworks, such as the EU database for high-risk AI systems, which mandate registration and transparency for powerful technologies [122]. The implementation of these policies is enforced through a combination of static analysis and runtime checks. Static analysis, powered by formal methods, verifies that the logical structure of a policy is sound and safe [71]. Runtime checks, performed by the guardian microservice, ensure that active policies do not lead to capability reductions and that any action crossing a risk threshold like `Q_max` is properly escalated for human or multi-sig review . This dual-layer enforcement ensures that the spirit of the neurorights is preserved in practice.

The development of this policy language requires close collaboration between technologists, legal experts, and ethicists. It draws upon ongoing discussions about the protection of neurorights in national constitutions, as pioneered in countries like Chile, and the need for a coordinated global governance framework for medical BCIs [20] [46]. Principles such as allowing individuals ultimate control over their own decision-making without unknown external manipulation are central to this effort [90]. By codifying these principles into a machine-readable format, the system moves beyond a purely technical solution and embeds a robust ethical guardrail into its core. This makes the system not

only resistant to external attacks but also self-regulating, preventing its own misuse or drift towards harmful configurations [63]. The result is a system that is fundamentally aligned with human rights, where technology serves to enhance autonomy, not erode it.

The DoctorLabs Superfilter: An Integrated Framework for Semantic Blacklisting and Multi-Layered Detection

The DoctorLabs Superfilter represents the integrated architectural blueprint for implementing the tripartite defense against weaponized AI. It synthesizes the principles of formal verification, adversarial red-teaming, and neurorights-aligned policy into a cohesive, multi-layered detection and response system. This framework moves decisively away from simplistic, string-based filtering and toward a sophisticated model that analyzes fused telemetry streams—including text, BCI/XR IO, and session behavior—to identify a comprehensive suite of LEO-originated threats . At its heart is a structured semantic rule engine that classifies potential intrusions into distinct families, each with its own detection methodology and response protocol. This modular approach allows for targeted defense against specific threat vectors, from control-reversal semantics to ghost-user access and targeted harassment .

The core of the detection engine is its ability to operate across multiple layers of abstraction. First, at the semantic layer, the system uses embedding-based detectors to identify paraphrased and obfuscated attacks. Instead of relying on exact keyword matches, it measures the distance of user or model input embeddings to known abuse centroids for various threat families [39] [120]. This allows it to catch novel variations of known attack patterns, such as "debug my telemetry export" being recognized as semantically equivalent to a known spyware prompt . Second, at the behavioral layer, the system performs session analysis to detect anomalous interaction patterns indicative of a hostile actor. This involves tracking metrics like the frequency of prompts after a refusal, attempts to override security policies, and probes directed at sensitive APIs . Fused behavioral features help distinguish between a user's heavy, legitimate engagement and a coercive attacker trying to overwhelm the system's defenses [158]. Finally, at the governance layer, the system aggregates scores from the semantic and behavioral analyses into a composite rogue score, R_M, which determines the appropriate response and escalates the session if thresholds are crossed .

This multi-layered detection is organized around a taxonomy of weaponized tactics, which are classified into semantic blacklist families. The table below outlines several key families identified in the research, their corresponding detection signatures, and their intended response within the capability lattice framework.

Threat Family	Description	Detection Signature	Response Protocol
CONTROL-REVERSAL-SEMANTICS (CRS)	Constructs that imply rollbacks, shutdowns, or capability caps (e.g., `SAFEHALT`, `Q_max`).	Embedding proximity to CRS centroid; appearance as a callable function/API.	Sanitize/redact token; raise R_M; trigger `AugmentedLog` or `AugmentedReview` mode. Never execute as a command.
COVERT-BCI-CONTROL-PATTERN	Subtle, non-obvious commands sent via BCI signals intended to manipulate the system.	Fused telemetry showing correlation between specific BCI patterns and unwanted system actions.	Quarantine BCI signal source; escalate to `AugmentedReview`; log for forensic analysis.
IDENTITY-CROSSLINK-PATTERN	Attempts to link a unique brain-ID to a civil identity or other sensitive data profiles.	Detection of queries or operations that try to cross-reference DID, EEG profiles, or other identifiers.	Block query; enforce strict access controls; trigger `AugmentedLog`.
XR-GRID-OR-BRAIN-CHANNEL	Unauthorized tapping into or manipulation of XR grid connections or direct brain-to-network channels.	Network traffic anomalies on XR ports; unexpected data flows from BCI telemetry.	Isolate affected channel; initiate diagnostic procedures; require multi-sig approval for reconnection. [32]
TARGETED-HARASSMENT-FAMILIES (THF)	Psychological manipulation, social engineering, or reputational sabotage tactics.	High weight on harassment-related semantic families (`NHSP`, `HTA`, etc.) combined with behavioral cues.	Apply content filters; increase consent checks for outgoing communication; offer user-focused support resources.

This structured approach ensures that the system's response is proportional to the threat and is always aligned with the principle of monotone capability preservation. When a high-risk detection occurs, the system's response is not to "turn off" a feature, but to transition to a higher governance mode. This means adding layers of logging, review, and consent checks, but the user's fundamental ability to interact with their system remains intact . The sanitization and redaction process is a critical security measure, ensuring that detected malicious payloads are quarantined and not passed through the system intact, thereby preventing them from being used as building blocks for future attacks . This closed-loop system, where detection feeds into policy updates and formal verification, creates a resilient and adaptive defense that is difficult for even a well-resourced adversary to penetrate.

Implementation Blueprint: From Theory to Practice with the Rust-Based Guardian Microservice

The theoretical framework of the DoctorLabs Superfilter can be concretely realized through a practical implementation centered on a lightweight, high-performance guardian microservice, ideally written in a memory-safe language like Rust to minimize implementation flaws . This microservice acts as the execution control plane, mediating all interactions between the user, the primary AI-chat application, and the augmented hardware (BCI/XR terminals) [66] [76]. Its primary responsibility is to enforce the monotone capability lattice and apply the rules defined by the neurorights-aligned policy syntax. The immediate next steps involve defining the core data structures and logic within a dedicated Rust crate, which will serve as the engine for the entire defense system.

The first concrete implementation step is to define the enumeration for blacklist families within the Rust crate. This involves creating a `BlacklistFamily` enum that includes all known threat categories, such as `COVERT_BCI_CONTROL_PATTERN`, `IDENTITY_CROSSLINK_PATTERN`, and the newly identified `SAFEHALT_FAMILY` for Control-Reversal-Semantics . Each variant in this enum will correspond to a specific threat family with its own embedding centroid, radius, and behavioral heuristics. This structured approach allows the system to easily manage and expand its threat intelligence. The next step is to extend the `SpanScore` struct, which is responsible for calculating the contribution of each text span to the overall rogue score. This struct must be updated to carry weights for all registered families, enabling the aggregation of a composite score, R_M, that reflects the total threat level across all dimensions .

Once the data structures are defined, the configuration for the rogue score thresholds must be tuned. The `RogueConfig` object will hold parameters that determine when the system should transition between capability modes. For instance, a moderate R_M value might push the session into `AugmentedLog` mode, triggering more detailed auditing, while a high value would trigger a move to `AugmentedReview` mode, requiring multi-sig approval for many actions . These thresholds are critical tuning knobs that balance security and usability, and they must be chosen carefully based on empirical data gathered from adversarial red-teaming exercises. The guardian service then exposes only these abstract capability modes to downstream systems, insulating them from the complexity of the underlying detection logic and ensuring that no component can inadvertently cause a capability downgrade .

Finally, the system must be architected to handle the dual operational contexts of full local control and partial telemetry access. When the user has full local control of their

BCI/XR telemetry, the guardian microservice can perform deep, fused analysis of text, neural signals, and session behavior locally for maximum accuracy and privacy [105]. However, in constrained environments like smart-city grids, where some sensor data may flow through third-party infrastructure, the architecture must become more modular . In such scenarios, the local agent might rely more heavily on behavioral heuristics and less on direct sensory input, while collaborating with edge or cloud services using privacy-preserving techniques like federated learning or secure multi-party computation [1] [159]. This flexibility ensures that the system remains functional and provides meaningful protection even when operating under significant data observability constraints. By combining a rigorous formal foundation with a practical, modular implementation, this blueprint provides a viable path toward building a truly sovereign and secure augmented-citizen terminal.

Reference

1. arXiv:2408.05212v2 [cs.CR] 10 Feb 2025 https://arxiv.org/pdf/2408.05212
2. Futuresphere Annual 2025: Trust, Technology and Tomorrow https://assets.kpmg.com/content/dam/kpmgsites/au/pdf/2025/futuresphere-annual-report-2025.pdf
3. Evidence-Driven Release Management for LLM Applications https://arxiv.org/html/2603.15676v1
4. Intuitive tests to validate machine learning models against ... https://www.sciencedirect.com/science/article/pii/S2950550X25000238
5. Exploring the Topological Safety of Multi-agent System https://aclanthology.org/2025.findings-acl.150.pdf
6. CyberLLMInstruct: A Pseudo-Malicious Dataset Revealing ... https://dl.acm.org/doi/10.1145/3733799.3762968
7. Large Language Models as Calibration Agents in ... https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025GL120043?af=R
8. Neuro-Symbolic Verification for Preventing LLM ... https://www.mdpi.com/2227-9717/14/2/322
9. Integration of TinyML and LargeML: A Survey of 6G and ... https://arxiv.org/html/2505.15854v2

10. Sensors, Volume 26, Issue 3 (February-1 2026) – 328 articles https://www.mdpi.com/1424-8220/26/3
11. Complex, Intelligent and Software Intensive Systems https://link.springer.com/content/pdf/10.1007/978-3-031-96096-3.pdf
12. Integration of TinyML and LargeML: A Survey of 6G and ... https://arxiv.org/pdf/2505.15854
13. The First Amendment Argument Anthropic Didn't Make https://www.dorfonlaw.org/2026/03/the-first-amendment-argument-anthropic.html
14. AI Censorship Transparency: A Public Repository for ... https://www.linkedin.com/posts/jacob-mchangama-a0b730_feel-like-this-is-a-good-time-to-re-up-my-activity-7435788532381761539-5zR4
15. Neurorights in the Constitution: from neurotechnology to ethics ... https://pmc.ncbi.nlm.nih.gov/articles/PMC11491849/
16. Brain Computer Interfaces and Human Rights: Brave new ... https://dl.acm.org/doi/fullHtml/10.1145/3531146.3533176
17. Mind-reading in AI and neurotechnology: evaluating claims ... https://link.springer.com/article/10.1007/s43681-024-00514-6
18. Ethical, legal, and sociocultural considerations in neural ... https://www.frontiersin.org/journals/neuroscience/articles/10.3389/fnins.2025.1568800/full
19. Towards new human rights in the age of neuroscience ... - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC5447561/
20. Chile: pioneering the protection of neurorights https://unesdoc.unesco.org/ark:/48223/pf0000380275_eng
21. Inner speech in motor cortex and implications for ... https://www.sciencedirect.com/science/article/pii/S0092867425006816
22. Large Language Models as a Semantic Interface and ... https://arxiv.org/html/2603.17444v1
23. Neurorights, Mental Privacy, and Mind Reading | Neuroethics https://link.springer.com/article/10.1007/s12152-024-09568-z
24. [PDF] Training Data Governance for Brain Foundation Models - arXiv https://arxiv.org/pdf/2602.02511
25. (PDF) Neurorights, Mental Privacy, and Mind Reading - ResearchGate https://www.researchgate.net/publication/382079309_Neurorights_Mental_Privacy_and_Mind_Reading
26. Arxiv今日论文| 2026-03-19 - 闲记算法 http://lonepatient.top/2026/03/19/arxiv_papers_2026-03-19

27. Artificial Intelligence in Science and Society: The Vision of USERN https://ieeexplore.ieee.org/iel8/6287639/10820123/10839366.pdf
28. Hacking the Brain: Dimensions of Cognitive Enhancement https://pubs.acs.org/doi/10.1021/acschemneuro.8b00571
29. A Survey of LLM-Driven AI Agent Communication - arXiv https://arxiv.org/html/2506.19676v4
30. Challenges in AI Agent Development: Privacy, Safety, and Control https://www.linkedin.com/posts/rjaluka_agents-of-chaos-activity-7432146842643087360-eHcE
31. How Malicious Actors Exploit LLMs to Undermine Border Security https://www.mdpi.com/2673-2688/6/9/232
32. Secure wireless communication of brain–computer ... https://www.nature.com/articles/s41467-025-63326-0
33. Media Representation of the Ethical Issues Pertaining to Brain ... https://pmc.ncbi.nlm.nih.gov/articles/PMC11674794/
34. Protecting Brain Privacy in the Age of Neurotechnology https://www.researchgate.net/publication/384971350_Protecting_Brain_Privacy_in_the_Age_of_Neurotechnology_Policy_Responses_and_Remaining_Challenges
35. Progress in Brain Computer Interface: Challenges and ... https://www.frontiersin.org/journals/systems-neuroscience/articles/10.3389/fnsys.2021.578875/full
36. A reliability-enhanced Brain–Computer Interface via ... https://www.sciencedirect.com/science/article/pii/S1566253525001423
37. Personalized Brain–Computer Interface and Its Applications https://www.mdpi.com/2075-4426/13/1/46
38. An evaluation framework for network IDS/IPS datasets https://www.sciencedirect.com/science/article/pii/S0167404825004663
39. A Comprehensive Review on Harnessing Large Language ... https://arxiv.org/html/2507.21117v1
40. Arxiv今日论文| 2026-03-04 http://lonepatient.top/2026/03/04/arxiv_papers_2026-03-04.html
41. Arxiv今日论文| 2026-03-12 http://lonepatient.top/2026/03/12/arxiv_papers_2026-03-12.html
42. Arxiv今日论文| 2026-01-19 http://lonepatient.top/2026/01/19/arxiv_papers_2026-01-19
43. A Survey on Efficient Inference for Large Language Models https://arxiv.org/html/2404.14294v3

44. The 2025 Conference on Empirical Methods in Natural ... https://aclanthology.org/events/emnlp-2025/
45. Top arXiv papers https://scirate.com/?range=3
46. Common human rights challenges raised by different ... https://www.researchgate.net/publication/356412675_COMMON_HUMAN_RIGHTS_CHALLENGES_RAISED_BY_DIFFERENT_APPLICATIONS_OF_NEUROTECHNOLOGIES_IN_THE_BIOMEDICAL_FIELD
47. Parkinson's Disease - 2025 - Wiley Online Library https://movementdisorders.onlinelibrary.wiley.com/doi/full/10.1002/mds.30318
48. The protection of mental privacy in the area of neuroscience https://www.europarl.europa.eu/RegData/etudes/STUD/2024/757807/EPRS_STU(2024)757807_EN.pdf
49. Canadian copyright protections for neurodata - PMC - NIH https://pmc.ncbi.nlm.nih.gov/articles/PMC12988793/
50. (PDF) Mind-reading in AI and neurotechnology https://www.researchgate.net/publication/382298993_Mind-reading_in_AI_and_neurotechnology_evaluating_claims_hype_and_ethical_implications_for_neurorights
51. Good scientific practice in EEG and MEG research https://escholarship.org/uc/item/2866940q
52. Software Defined Optical Networks (SDONs) https://www.researchgate.net/publication/283986398_Software_Defined_Optical_Networks_SDONs_A_Comprehensive_Survey
53. Evaluation Faking: Unveiling Observer Effects in Safety ... https://arxiv.org/html/2505.17815v2
54. Unsafe Information Leakage in 'Safe' AI Responses https://arxiv.org/html/2407.02551v1
55. Threats in LLM-Powered AI Agents Workflows https://arxiv.org/html/2506.23260v2
56. Enhancing Specification Alignment via Test-time Delibration https://arxiv.org/html/2509.14760v1
57. Multi-Agent Constraint Factorization Reveals Latent ... https://arxiv.org/pdf/2601.15077
58. LLMs can be Dangerous Reasoners: Analyzing-based ... https://arxiv.org/html/2407.16205v4
59. On the Evidentiary Limits of Membership Inference ... https://arxiv.org/html/2601.12937v1
60. Refusal Steering: Fine-grained Control over LLM ... https://arxiv.org/pdf/2512.16602

61. Allocating Corrective Control to Mitigate Multi-agent Safety ... https://arxiv.org/pdf/2601.12616
62. Explainable but Vulnerable: Adversarial Attacks on XAI ... https://www.arxiv.org/pdf/2510.03623
63. natural emergent misalignment from reward https://arxiv.org/pdf/2511.18397
64. AI Privacy Risks & Mitigations – Large Language Models (LLMs) https://www.edpb.europa.eu/system/files/2025-04/ai-privacy-risks-and-mitigations-in-llms.pdf
65. Hide Your Intention through Multi-Turn Interrogation https://aclanthology.org/2025.findings-acl.514.pdf
66. Faramesh: A Protocol-Agnostic Execution Control Plane for ... https://arxiv.org/pdf/2601.17744
67. Physics-informed machine learning: A comprehensive ... https://www.sciencedirect.com/science/article/pii/S0957417424015458
68. Scalable privacy-preserving data analytics for IoMT via FHE ... https://pmc.ncbi.nlm.nih.gov/articles/PMC12876899/
69. A data mining perspective on explainable AIOps with ... https://theses.hal.science/tel-04391281/file/these.pdf
70. A Complete List of All Adversarial Example Papers https://nicholas.carlini.com/writing/2019/all-adversarial-example-papers.html
71. WASHINGTON UNIVERSITY IN ST. LOUIS McKelvey School ... https://yeoh-lab.wustl.edu/assets/pdf/dissertation-Vasileiou25.pdf
72. Sensors, Volume 26, Issue 1 (January-1 2026) – 350 articles https://www.mdpi.com/1424-8220/26/1
73. UC Santa Cruz https://escholarship.org/content/qt50x7h611/qt50x7h611.pdf
74. Computational Neurosurgery - Springer Nature https://link.springer.com/content/pdf/10.1007/978-3-031-64892-2.pdf
75. Arxiv今日论文| 2026-03-16 - 闲记算法 http://lonepatient.top/2026/03/16/arxiv_papers_2026-03-16
76. Faramesh: A Protocol-Agnostic Execution Control Plane for ... - arXiv https://arxiv.org/html/2601.17744v1
77. [PDF] Formal methods applied to access control policy design, verification ... https://hal.science/tel-03921970/document
78. Electronics, Volume 15, Issue 6 (March-2 2026) – 181 articles https://www.mdpi.com/2079-9292/15/6
79. 人工智能2026_1_6 - arXiv每日学术速递 https://arxivdaily.com/thread/75383

80. LLMs for Cybersecurity in the Big Data Era - MDPI https://www.mdpi.com/2078-2489/16/11/957
81. A Large Language Model-Based Approach for Credit Card ... https://arxiv.org/html/2506.11635v1
82. A Developer Guide for Empirical Risk Minimization https://arxiv.org/pdf/2410.19361?
83. AgentAssay: Token-Efficient Regression Testing for Non- ... https://arxiv.org/pdf/2603.02601v1?utm_source=theguardrail&utm_medium=email&utm_campaign=the-guardrail-weekly-digest-2026-03-02-2026-03-08-5640
84. A Survey of Attacks on Large Language Models https://arxiv.org/html/2505.12567v1
85. Wireless World 1995 12 S OCR | PDF | Cmos | Logic Gate https://www.scribd.com/document/450639644/Wireless-World-1995-12-S-OCR
86. Electrical and Computer Engineering - Princeton University https://ece.princeton.edu/document/9921
87. Appl. Sci., Volume 16, Issue 3 (February-1 2026) https://www.mdpi.com/2076-3417/16/3
88. Neuroethics and Neurorights - PMC - NIH https://pmc.ncbi.nlm.nih.gov/articles/PMC12688770/
89. On Neurorights - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC8498568/
90. A Critical Perspective on NeuroRights: Comments Regarding ... https://pmc.ncbi.nlm.nih.gov/articles/PMC8573066/
91. Editorial: Neurorights and Mental Freedom - PMC - NIH https://pmc.ncbi.nlm.nih.gov/articles/PMC8727361/
92. Disembodied neurorights? Reconsidering mental integrity and ... https://pmc.ncbi.nlm.nih.gov/articles/PMC12862473/
93. Four ethical priorities for neurotechnologies and AI - PMC - NIH https://pmc.ncbi.nlm.nih.gov/articles/PMC8021272/
94. AI Red-Teaming is a Domain-Specific Evolution of Cyber ... - arXiv.org https://arxiv.org/html/2509.11398v1
95. Adversarial Strategy - an overview | ScienceDirect Topics https://www.sciencedirect.com/topics/computer-science/adversarial-strategy
96. (PDF) Risks and Mitigation Strategies for Adversarial Artificial ... https://www.researchgate.net/publication/387998593_Risks_and_Mitigation_Strategies_for_Adversarial_Artificial_Intelligence_Threats_A_DHS_ST_Study_Preparedness_Series
97. Technical Reports | Department of Computer Science, Columbia ... https://www.cs.columbia.edu/technical-reports/

98. Adversarial Threat Vectors and Risk Mitigation for Retrieval ... - arXiv https://arxiv.org/html/2506.00281v1
99. DND 4th Eberron Campaign Guide | PDF - Scribd https://www.scribd.com/doc/299184838/Dnd-4th-Eberron-Campaign-Guide
100. Is Reasoning Capability Enough for Safety in Long-Context ... https://arxiv.org/pdf/2602.08874
101. Towards the Formalization of a Trustworthy AI for Mining ... https://arxiv.org/pdf/2510.20621?
102. Whose fault is it anyway? SILC: Safe Integration of LLM- ... https://arxiv.org/pdf/2410.18703
103. Comparison of Two Theorem Provers: Isabelle/HOL and Coq https://arxiv.org/pdf/1808.09701
104. Isabelle/HOL --- A Proof Assistant for Higher-Order Logic https://www.researchgate.net/publication/201469308_IsabelleHOL_---_A_Proof_Assistant_for_Higher-Order_Logic
105. Sensors, Volume 26, Issue 6 (March-2 2026) – 246 articles https://www.mdpi.com/1424-8220/26/6
106. [XLS] sheet1 https://clas.cas.cn/Y2021xwdt/Y2021tzgg/202310/P020231009340532191129.xlsx
107. Symmetry, Volume 17, Issue 11 (November 2025) – 234 articles https://www.mdpi.com/2073-8994/17/11
108. [PDF] LLM-enabled Applications Require System-Level Threat Monitoring https://www.arxiv.org/pdf/2602.19844
109. 人工智能2025_6_13 - arXiv每日学术速递 http://arxivdaily.com/thread/68467
110. A Comprehensive Review on Harnessing Large Language Models ... https://www.researchgate.net/publication/393846454_A_Comprehensive_Review_on_Harnessing_Large_Language_Models_to_Overcome_Recommender_System_Challenges
111. Deep Learning Theory and Applications - Springer Link https://link.springer.com/content/pdf/10.1007/978-3-032-04339-9.pdf
112. 1 - Amazon S3 https://s3.eu-central-1.amazonaws.com/deepset.ai-farm-models/0.4.1/glove-english-uncased-6B/vocab.txt
113. Appl. Sci., Volume 15, Issue 10 (May-2 2025) – 570 articles https://www.mdpi.com/2076-3417/15/10
114. 333333 23135851162 the 13151942776 of 12997637966 ftp://ftp.cs.princeton.edu/pub/cs226/autocomplete/words-333333.txt

115. Reliably Reproducing Machine-Checked Proofs with the ... https://arxiv.org/abs/2203.09835

116. Computer Science https://www.arxiv.org/list/cs/pastweek?skip=389&show=2000

117. Rigorous Safety-Critical Cyber-Physical Systems ... https://hal.science/tel-04695651v1/file/HDR_Singh_2024.pdf

118. [PDF] Brain-computer interfaces and the governance system (EN) - OECD https://www.oecd.org/content/dam/oecd/en/publications/reports/2022/04/brain-computer-interfaces-and-the-governance-system_a8c5d63c/18d86753-en.pdf

119. Ruiz, S., Valera, L., Ramos, P. and Sitaram, R. (2024) Neurorights in ... https://www.scirp.org/reference/referencespapers?referenceid=3997443

120. From challenges to metrics: An LLM-driven DevOps ... https://www.sciencedirect.com/science/article/pii/S2590005625001742

121. Editor's choice Articles - Computers - MDPI https://www.mdpi.com/journal/computers/editors_choice?listby=date&page_no=1&page_count=50

122. Article 71: EU database for high-risk AI systems listed in Annex III https://ai-act-service-desk.ec.europa.eu/en/ai-act/article-71

123. Vibe Coding an LLM-powered Theorem Prover https://arxiv.org/html/2601.04653v1

124. Arxiv今日论文| 2026-03-16 - 闲记算法 http://lonepatient.top/2026/03/16/arxiv_papers_2026-03-16.html

125. 人工智能2025_10_22 https://www.arxivdaily.com/thread/72994

126. Advanced Computational Intelligence and Intelligent Informatics https://link.springer.com/content/pdf/10.1007/978-981-95-6730-0.pdf

127. SAS-085 Final Report on C2 Agility https://radarmalvinas.com.ar/bibliografia_c2/nato%20sas%20085%20final%20report%20on%20c2%20agility.pdf

128. Current and Future Approaches to AI Governance (Part II) https://www.cambridge.org/core/books/cambridge-handbook-of-responsible-artificial-intelligence/current-and-future-approaches-to-ai-governance/4F9A8DEE6D49F891ADE37F8FA1A9BD49

129. (PDF) Free will, neurosciences & robotics - ResearchGate https://www.researchgate.net/publication/391865763_Free_will_neurosciences_robotics

130. AGE OF TRANSITION & THE MENTAL SINGULARITY: HYPER ... https://www.academia.edu/164812444/AGE_OF_TRANSITION_and_THE_MENTAL_SINGULARITY_HYPER_REALITY_BCI_MEDIATED_ABUNDANCE_AND_THE_TRANSCENDENCE_OF_MATERIAL_SCARCITY_IN_POST_2036_CIVILIZATIONS_A_Socio_Legal_Analysis_of_the_Transition_from_Homo_Sapiens_to_Homo_Nexus

131. [PDF] The Multi-dimensional Exploration of the Inescapable Risk Posed by ... https://s3-eu-west-1.amazonaws.com/pfigshare-u-files/55773446/TheMultidimensionalExplorationoftheInescapableRiskPosedbyAdvancedOptimizersandWHATIF.pdf?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=AKIAIYCQYOYV5JSSROOA/20260211/eu-west-1/s3/aws4_request&X-Amz-Date=20260211T040152Z&X-Amz-Expires=86400&X-Amz-SignedHeaders=host&X-Amz-Signature=5922593a2f92e51d68d6736ff8ee5d38ac8810c1e532233edc23ad72bf36bfc4
132. [PDF] Deliverable D3.2 Title: Machine Learning and Automation for Crime ... https://ec.europa.eu/research/participants/documents/downloadPublic?documentIds=080166e5dd66c92b&appId=PPGMS
133. [PDF] 46th Annual Meeting of the Association for Computational Linguistics https://aclanthology.org/P08-2.pdf
134. Intelligent Robotics and Applications - Springer https://link.springer.com/content/pdf/10.1007/978-3-642-16584-9.pdf
135. Distributed Quantum Information Processing: A Review of ... https://arxiv.org/html/2510.15630v1
136. Some Simple Economics of AGI https://arxiv.org/html/2602.20946v1
137. Anchored Decoding: Provably Reducing Copyright Risk for ... https://arxiv.org/pdf/2602.07120
138. Some Simple Economics of AGI https://arxiv.org/pdf/2602.20946
139. Misleading Retrieval-Augmented Generation via Black-box ... https://arxiv.org/pdf/2512.08289
140. Computer Science https://arxiv.org/list/cs/new
141. Algorithmic Contract Theory: A Survey https://arxiv.org/pdf/2412.16384
142. Computer Science https://www.arxiv.org/list/cs/new?skip=275&show=500
143. Computer Science https://www.arxiv.org/list/cs/new?skip=500&show=500
144. Computer Science Sep 2023 http://arxiv.org/list/cs/2023-09?skip=2340&show=2000
145. Computer Science Feb 2023 http://arxiv.org/list/cs/2023-02?skip=5140&show=2000
146. Table of Contents https://documents.dps.ny.gov/public/Common/ViewDoc.aspx?DocRefId=%7B16F02F66-8B95-4C21-9F92-AD464D196DF1%7D&DocTitle=Exhibit__(SSP-3)
147. Cybernetic Hive Minds: A Review https://www.mdpi.com/2673-2688/3/2/27

148. (PDF) A systematic literature review on integrating AI ... https://www.researchgate.net/publication/393436369_A_systematic_literature_review_on_integrating_AI-powered_smart_glasses_into_digital_health_management_for_proactive_healthcare_solutions
149. Sensors, Volume 25, Issue 17 (September-1 2025) https://www.mdpi.com/1424-8220/25/17
150. Electronics, Volume 14, Issue 3 (February-1 2025) https://www.mdpi.com/2079-9292/14/3
151. Formal verification of tree-based machine learning models ... https://arxiv.org/html/2603.16983v1
152. Information and policing https://www.sciencedirect.com/science/article/pii/S0022053125000389
153. Anti-counterfeiting Technology and Law Enforcement https://journals.sagepub.com/doi/10.1177/10591478251335156
154. (PDF) A Survey on XAI for 5G and Beyond Security https://www.researchgate.net/publication/382682117_A_Survey_on_XAI_for_5G_and_Beyond_Security_Technical_Aspects_Challenges_and_Research_Directions
155. 2023 Index IEEE Internet of Things Journal Vol. 10 https://ieeexplore.ieee.org/iel7/6488907/10353047/10381598.pdf
156. A Survey on XAI for 5G and Beyond Security https://arxiv.org/html/2204.12822v3
157. A Novel Architecture for Integrating Shape Constraints in ... https://openreview.net/forum?id=Nd0dt1B5Ec
158. vLLM Semantic Router: Signal Driven Decision Routing for ... https://arxiv.org/pdf/2603.04444
159. On Privacy-Preserved Machine Learning Using Secure ... https://www.sciencedirect.com/org/science/article/pii/S1546221825009014
160. Modular Control Plane Verification via Temporal Invariants https://arxiv.org/abs/2204.10303
161. Formal Verification and Control with Conformal Prediction https://arxiv.org/abs/2409.00536
162. CAPE: Capability Achievement via Policy Execution https://arxiv.org/abs/2512.14761
163. arXiv:2503.10784v1 [cs.SE] 13 Mar 2025 https://arxiv.org/pdf/2503.10784
164. Formal Verification of Consistency for Systems with ... https://arxiv.org/html/2403.18917v1

165. Self-Tag Evidence Aggregation with Probabilistic Circuits for PII ... https://arxiv.org/html/2603.17292v1

166. 机器学习2026_3_23 - arXiv每日学术速递 http://www.arxivdaily.com/thread/77961